Saehun Chun

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RESEARCH INTERESTS

- Robot Foundation Models and Embodied AI
- Efficient AI Systems and Multimodal Model Serving
- Memory and Computation Reuse for Interactive AI

EDUCATION

2025–Present **Sungkyunkwan University** Suwon, Republic of Korea
Integrated M.S./Ph.D. Program in Software
Department of Software | Lab: CSI Lab | Advisor: Honguk Woo
GPA: 4.14 / 4.5

2019–2024 **Sungkyunkwan University** Suwon, Republic of Korea
B.S. in Software
Department of Software | GPA: 3.41 / 4.5

RESEARCH EXPERIENCE

2024–Present **CSI Lab, Sungkyunkwan University** Suwon, Republic of Korea
Graduate Researcher

- Study efficient VLA and multimodal inference, focusing on temporal reuse, asynchronous execution, and stale-state mitigation.
- Develop structured memory and KV reuse for embodied agents (HeMeR, NeurIPS 2026 Oral).
- Develop function-level computation reuse for embodied code policies (FCGraft, ICML 2026 Poster).

2023–2024 **VLDB Lab, Sungkyunkwan University** Suwon, Republic of Korea
Undergraduate Research Assistant

- Investigated undo-page buffer pollution in HTAP workloads and evaluated memory-management mechanisms in MySQL/InnoDB using C++.

PUBLICATIONS

5 PAPERS

International Conferences (3)

[1] HeMeR: Heterogeneous Memory Reconciliation for Embodied Agents via Structured KV Reuse.
Saehun Chun, Wonje Choi, Jinwoo Jang, TaeYoon Kwack, and Honguk Woo*.
Conference on Neural Information Processing Systems (NeurIPS), 2026. **Oral.**

Reconciles declarative, procedural, and working memory through selective KV-state reuse. On real-world tasks, it raises success from 64.28% to 83.20% and reduces inference time from 2.94 to 1.87 s/step versus MemRL.

[2] Functional Cache Grafting: Robust and Rapid Code-Policy Synthesis for Embodied Agents.
Saehun Chun, Wonje Choi, Sera Choi, Sanghyun Ahn, and Honguk Woo*.
International Conference on Machine Learning (ICML), 2026. **Poster.**

Composes validated robot-policy functions through cache-stitching and repairs faulty spans through cache-patching. It achieves an average 2.3× speedup in policy synthesis over RAGCache across the evaluated tasks.

[3] Efficient Skill Grounding via Code Refactoring with Small Language Models.
Sera Choi, Wonje Choi, **Saehun Chun**, Daehee Lee, Chaeun Lee, Jooyoung Kim, and Honguk Woo*.
International Conference on Machine Learning (ICML), 2026. **Poster.**

Grounds reusable skills through ontology-guided, localized code refactoring while preserving skill semantics. RECENT achieves 73.0% and 82.5% task success under kinematic and end-effector variations, respectively.

Domestic Conferences (2)

[1] Undo Read Buffer: Preventing Buffer Pollution due to Undo Blocks in HTAP Workloads.
Saehun Chun, Jonghyeok Park, Sang-Won Lee*, and Young Ik Eom.
Korea Computer Congress (KCC), 2024.

[2] Analysis of Buffer Pollution Caused by Undo Pages in HTAP Workloads.
Saehun Chun, Bo-Hyun Lee, Kyong-Shik Lee, Sang-Won Lee*, and Young Ik Eom.
Korean DataBase Conference (KDBC), 2023.

RESEARCH PROJECTS

2024–Present **HiPER: High-Performance Exabyte Storage Systems**
Project Manager / Graduate Researcher
Samsung Electronics

- Develop and benchmark memory- and storage-efficient support for foundation-model and embodied-agent workloads.
- Coordinate experiments, prototypes, and technical deliverables for the Samsung Electronics collaboration.

Jan. 2024 **Global Capstone Design**
Project Manager
Sungkyunkwan University & Victoria University of Wellington

- Prototyped ML-based MySQL buffer management; presented the work and collaborated with students at Victoria University of Wellington.

2020 **Industry-Academic Collaboration Project**
Project Manager
SK M&N Service

- Led development of an STT-based video summarization system using Python and TensorFlow. Repository.

AWARDS AND HONORS

2025–2026 **Graduate Research Fellowship**, National Research Foundation of Korea.

TEACHING EXPERIENCE

2026–Present **GenAI PowerUser Program Level 4 (Multi Agent framework), Samsung Electronics University**
Hands-on Training Instructor

Fall 2026 **Artificial Intelligence Project, Sungkyunkwan University**
Hands-on Training Instructor

Spring 2026 **System Programming, Sungkyunkwan University**
Teaching Assistant

Spring 2025 **Mobile Application Programming, Sungkyunkwan University**
Teaching Assistant

RESEARCH EXPERTISE AND TECHNICAL SKILLS

Embodied AI and Skill Grounding: Foundation-model agents, executable robot policies, ontology-guided skill adaptation, and feedback-driven policy repair.

Efficient Inference: Computation reuse for VLA and multimodal models, asynchronous inference, and stale-state mitigation.

Agent Memory: Declarative, procedural, and working memory; LoRA adapter-aware KV reuse and chunk-level state reconciliation.

ML Systems: Memory- and storage-efficient model serving, cache management, and HTAP buffer management.

Benchmarks and Platforms: ALFWorld, Habitat, ALFRED, TEACh, RLBench; Franka Research 3 (FR3) and UR7e.